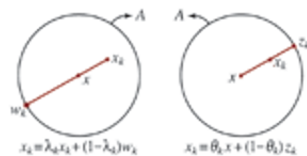
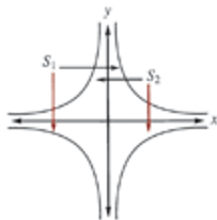
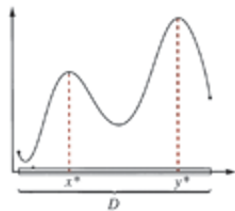
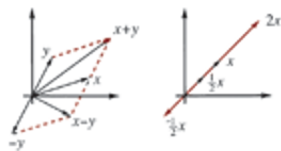


A First Course in Optimization Theory



Rangarajan K. Sundaram

A FIRST COURSE IN OPTIMIZATION THEORY

A First Course in Optimization Theory introduces students to optimization theory and its use in economics and allied disciplines. The first of its three parts examines the existence of solutions to optimization problems in $\mathbb{R}^n$, and how these solutions may be identified. The second part explores how solutions to optimization problems change with changes in the underlying parameters, and the last part provides an extensive description of the fundamental principles of finite- and infinite-horizon dynamic programming.

Each chapter contains a number of detailed examples explaining both the theory and its applications for first-year master's and graduate students. "Cookbook" procedures are accompanied by a discussion of when such methods are guaranteed to be successful, and equally importantly, when they could fail. Each result in the main body of the text is also accompanied by a complete proof. A preliminary chapter and three appendices are designed to keep the book mathematically self-contained.

A FIRST COURSE
IN OPTIMIZATION THEORY

RANGARAJAN K. SUNDARAM
New York University



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To
my parents
and
to my grandmother Innumma

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Preface

This book developed out of a course I have taught since 1988 to first-year Ph.D. students at the University of Rochester on the use of optimization techniques in economic analysis. A detailed account of its contents is presented in Section 2.5 of Chapter 2. The discussion below is aimed at providing a broad overview of the book, as well as at emphasizing some of its special features.

An Overview of the Contents

The main body of this book may be divided into three parts. The first part, encompassing Chapters 3 through 8, studies optimization in n -dimensional Euclidean space, $\mathbb{R}^n$. Several topics are covered in this span. These include—but are not limited to—(i) the Weierstrass Theorem, and the existence of solutions to optimization problems; (ii) the Theorem of Lagrange, and necessary conditions for optima in problems with *equality* constraints; (iii) the Theorem of Kuhn and Tucker, and necessary conditions for optima in problems with *inequality* constraints; (iv) the role of *convexity* in obtaining *sufficient* conditions for optima in constrained optimization problems; and (v) the extent to which convexity can be replaced with *quasi*-convexity, while still obtaining sufficiency of the first-order conditions for global optima.

The second part of the book, comprised of Chapters 9 and 10, looks at the issue of *parametric variation* in optimization problems, that is, at the manner in which solutions to optimization problems respond to changes in the values of underlying parameters. Chapter 9 begins this exercise with the question of *parametric continuity*: under what conditions will solutions to optimization problems vary “continuously” with changes in the underlying parameters? An answer is provided in the centerpiece of this chapter, the Maximum Theorem. The strengthening of the Maximum Theorem that is obtained by adding convexity restrictions to the problem is also examined. Chapter 10 is concerned with *parametric monotonicity*, that is, with conditions under which increases in the values of parameters result in increases in the size of optimal

actions. The important concept of *supermodularity*, and its role in securing parametric monotonicity, is described here.

The third and final part of the book, consisting of Chapters 11 and 12, explores optimization theory in a *dynamic* environment. Chapter 11 discusses dynamic optimization in problems with a *finite* horizon. It establishes the existence of optimal strategies under suitable conditions, and details the method of *backwards induction* by which these strategies may be recovered. Chapter 12 focuses on the infinite-horizon problem, and derives the Bellman optimality equation for this case. The role of this equation in identifying optimal strategies is discussed, and the existence of optimal strategies is established under suitable conditions.

Prerequisites

In principle, this book is mathematically self-contained. There are four chapters (Chapter 1, and Appendices A, B, and C) devoted exclusively to providing the reader with a description of the mathematical results used in the body of the text. Of these, Appendix A is concerned with the basic rules of logic; among other things, it discusses the concept of *negation* and the importance of the order of quantifiers in forming a negation; the distinction between a *contrapositive* and a *converse*; and that between a *necessary* condition and a *sufficient* condition. It is vital that the reader have a mastery over this material before reading the rest of the text.

Chapter 1 presents a compendium of results from mathematical analysis and linear algebra that are used in the rest of the book. Strictly speaking, not all the contents of this chapter are required for a comprehension of the material that follows; rather, it suffices that the reader is familiar with a few topics such as continuous and differentiable functions on $\mathbb{R}^n$, closed, open, compact, and convex sets in $\mathbb{R}^n$, and definite and semidefinite matrices. However, the contents of Chapter 1 are invoked repeatedly in various *proofs* in the main body of the text. As such, a systematic knowledge of this material should significantly enhance understanding in the sequel.

It is presumed in Chapter 1 that the reader has at least an intuitive feel for what is meant by a “real number.” Appendix B, which describes the construction of the real line from rational numbers, is aimed at readers who either lack this knowledge, or who wish to have a somewhat more formal introduction to the real number line. However, it is to be emphasized that Appendix B is genuinely only an “appendix”; a knowledge of its contents is by no means required at any point in the book.

The presentation in Chapter 1 also takes place solely in the context of Euclidean spaces, although many of the structures presented here (inner product, norm, metric, etc.) can be developed in more abstract vector spaces. Appendix C is aimed at readers who wish to see the outline of such a development. It defines abstract vector spaces,

and describes four constructs on such spaces: inner product spaces, normed spaces, metric spaces, and topological spaces.

What Is Different About This Book?

In part, at least, this book owes its existence to a feeling of dissatisfaction with the available texts in this area. Many of the available texts concentrate on applications to the exclusion of theory. The cookbook approach they espouse is problematic for many reasons, not least because students do not obtain a clear understanding of even the assumptions underlying a particular result, let alone why those assumptions are required. At the other end of the spectrum, there are a number of books that are mathematically precise and detailed, but that do not elaborate on the use of the theory in *solving* optimization problems.

This book represents an attempt to avoid either of these pitfalls. The presentation has five features that aim at blending the theory with its applications. The first two features are directed at clarifying the theory itself; the next two concern the use of the theory in identifying solutions to given optimization problems; and the last involves *working* with both the theory and its application.

First, every result in the main body of the book is accompanied by a complete and detailed proof. To avoid disrupting the flow of material, these proofs are usually placed at the end of each chapter.

Second, following the statement of each important result, a number of examples are given that illustrate why the result would be invalid were any of the assumptions to be violated. Taken together with the proofs of the various results, these examples are designed to make the student appreciate the part played by each assumption in the overall result.

Third, each important result is also followed by a section discussing procedures for using the result in applications to actually locate an optimum. Where the results concerned are *sufficient* conditions, the discussion is relatively straightforward. However, where *necessary* conditions are involved (as in the Lagrange and Kuhn–Tucker Theorems), the “cookbook” procedures for using these results are not guaranteed to be uniformly successful. Consequently, the description of these procedures is also accompanied by a detailed discussion of when the procedure is guaranteed to be successful, and equally importantly, when it could fail.

Fourth, in each chapter, the use of the chapter’s results in applications is illustrated through detailed worked-out examples based on well-known paradigms. Thus, for instance, the utility-maximization problem from consumer theory, and the cost-minimization problem from producer theory, are each used to describe the use of Lagrange’s Theorem in solving equality-constrained optimization problems, while the two examples used to illustrate the use of the Maximum Theorem (and the

Maximum Theorem under Convexity) are taken from consumer theory, and game theory, respectively.

Finally, since the only real way to learn mathematics is to use it, there is a list of exercises provided at the end of each chapter. The exercises range from the very easy to the quite difficult. Some of the exercises emphasize the theory by, for example, requiring the students to find counterexamples to given statements. Others focus on applications, and ask students to detail optimization problems describing specific situations, and to solve them for given parametrizations.

Acknowledgements

No work of authorship is ever the accomplishment of one individual, and the present volume is no exception. It is a pleasure to be able to acknowledge the many people whose help and advice shaped this book.

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Karl Shell provided a great deal of support and encouragement throughout this project, most notably in its early stages when I was still doubtful of the wisdom of continuing with it. In this context, I am also grateful to Beth Allen, Ron Jones, Mukul Majumdar, Andy McLennan, and especially Marti Subrahmanyam.

As chairman of the Department of Economics at the University of Rochester, Alan Stockman enabled me to alter my teaching schedule and get two valuable months off in the Fall of 1994. With Jess Benhabib's assistance, I spent these two months at the Department of Economics at New York University, and it was in this period that over a third of the manuscript was written. I am indebted to Alan and Jess for their respective roles in this regard.

Scott Parris and the rest of the Cambridge University Press team (especially Linda Johnson and Janis Bolster) were always helpful and prompt. It was a pleasure working with them.

Perhaps my greatest debt is to my wife, Urmilla, who bore the many late nights and weekends of my work on this manuscript with patience, and whose support on the home front was what, in the final analysis, made completion of this project possible.

1

Mathematical Preliminaries

This chapter lays the mathematical foundation for the study of optimization that occupies the rest of this book. It focuses on three main topics: the topological structure of Euclidean spaces, continuous and differentiable functions on Euclidean spaces and their properties, and matrices and quadratic forms. Readers familiar with real analysis at the level of Rudin (1976) or Bartle (1964), and with matrix algebra at the level of Munkres (1964) or Johnston (1984, Chapter 4), will find this chapter useful primarily as a refresher; for others, a systematic knowledge of its contents should significantly enhance understanding of the material to follow.

Since this is not a book in introductory analysis or linear algebra, the presentation in this chapter cannot be as comprehensive or as leisurely as one might desire. The results stated here have been chosen with an eye to their usefulness towards the book's main purpose, which is to develop a theory of optimization in Euclidean spaces. The selective presentation of proofs in this chapter reveals a similar bias. Proofs whose formal structure bears some resemblance to those encountered in the main body of the text are spelt out in detail; others are omitted altogether, and the reader is given the choice of either accepting the concerned results on faith or consulting the more primary sources listed alongside the result.

It would be inaccurate to say that this chapter does not presuppose any knowledge on the part of the reader, but it is true that it does not presuppose much. Appendices A and B aim to fill in the gaps and make the book largely self-contained. Appendix A reviews the basic rules of propositional logic; it is taken for granted throughout that the reader is familiar with this material. An intuitive understanding of the concept of an “irrational number,” and of the relationship between rational and irrational numbers, suffices for this chapter and for the rest of this book. A formal knowledge of the real line and its properties will, however, be an obvious advantage, and readers who wish to acquaint themselves with this material may consult Appendix B.

The discussion in this chapter takes place solely in the context of Euclidean spaces. This is entirely adequate for our purposes, and avoids generality that we do not need. However, Euclidean spaces are somewhat special in that many of their properties (such as completeness, or the compactness of closed and bounded sets) do not carry over to more general metric or topological spaces. Readers wishing to view the topological structure of Euclidean spaces in a more abstract context can, at a first pass, consult Appendix C, where the concepts of inner product, norm, metric, and topology are defined on general vector spaces, and some of their properties are reviewed.

1.1 Notation and Preliminary Definitions

1.1.1 Integers, Rationals, Reals, $\mathbb{R}^n$

The notation we adopt is largely standard. The set of positive integers is denoted by $\mathbb{N}$, and the set of all integers by $\mathbb{Z}$:

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\} \\ \mathbb{Z} &= \{\dots, -2, -1, 0, 1, 2, \dots\}.\end{aligned}$$

The set of rational numbers is denoted by $\mathbb{Q}$:

$$\mathbb{Q} = \left\{ x \mid x = \frac{p}{q}, p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Finally, the set of all real numbers, both rational and irrational, is denoted by $\mathbb{R}$. As mentioned earlier, it is presumed that the reader has at least an intuitive understanding of the real line and its properties. Readers lacking this knowledge should first review Appendix B.

Given a real number $z \in \mathbb{R}$, its *absolute value* will be denoted $|z|$:

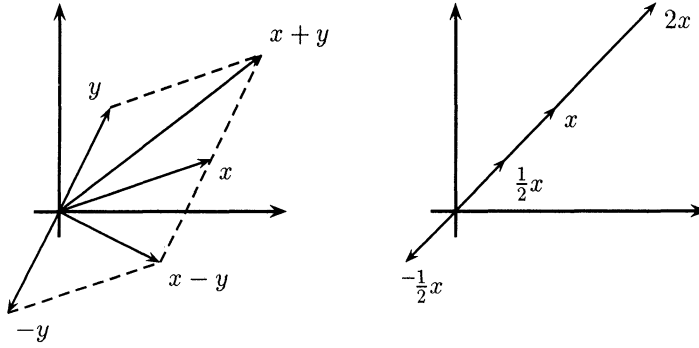
$$|z| = \begin{cases} z & \text{if } z \geq 0 \\ -z & \text{if } z < 0. \end{cases}$$

The *Euclidean distance* between two points x and y in $\mathbb{R}$ is defined as $|x - y|$, i.e., as the absolute value of their difference.

For any positive integer $n \in \mathbb{N}$, the n -fold Cartesian product of $\mathbb{R}$ will be denoted $\mathbb{R}^n$. We will refer to $\mathbb{R}^n$ as *n -dimensional Euclidean space*. When $n = 1$, we shall continue writing $\mathbb{R}$ for $\mathbb{R}^1$.

A point in $\mathbb{R}^n$ is a vector $x = (x_1, \dots, x_n)$ where for each $i = 1, \dots, n$, x_i is a real number. The number x_i is called the *i -th coordinate* of the vector x .

We use 0 to denote the real number 0 as well as the null vector $(0, \dots, 0) \in \mathbb{R}^n$. This notation is ambiguous, but the correct meaning will usually be clear from the context.

Fig. 1.1. Vector Addition and Scalar Multiplication in $\mathbb{R}^2$

Vector addition and scalar multiplication are defined in $\mathbb{R}^n$ as follows: for $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$,

$$x + y = (x_1 + y_1, \dots, x_n + y_n)$$

$$\alpha x = (\alpha x_1, \dots, \alpha x_n).$$

Figure 1.1 provides a graphical interpretation of vector addition and scalar multiplication in $\mathbb{R}^2$.

Given any two n -vectors $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$, we write

$$x = y, \text{ if } x_i = y_i, \quad i = 1, \dots, n.$$

$$x \geq y, \text{ if } x_i \geq y_i, \quad i = 1, \dots, n.$$

$$x > y, \text{ if } x \geq y \text{ and } x \neq y.$$

$$x \gg y, \text{ if } x_i > y_i, \quad i = 1, \dots, n.$$

Note that

- $x \geq y$ does *not* preclude the possibility that $x = y$, and
- for $n > 1$, the vectors x and y need not be comparable under any of the categories above; for instance, the vectors $x = (2, 1)$ and $y = (1, 2)$ in $\mathbb{R}^2$ do not satisfy $x \geq y$, but neither is it true that $y \geq x$.

The *nonnegative* and *strictly positive* orthants of $\mathbb{R}^n$, denoted $\mathbb{R}_+^n$ and $\mathbb{R}_{++}^n$, respectively, are defined as

$$\mathbb{R}_+^n = \{x \in \mathbb{R}^n \mid x \geq 0\},$$

and

$$\mathbb{R}_{++}^n = \{x \in \mathbb{R}^n \mid x \gg 0\}.$$

1.1.2 Inner Product, Norm, Metric

This subsection describes three structures on the space $\mathbb{R}^n$: the *Euclidean inner product* of two vectors x and y in $\mathbb{R}^n$, the *Euclidean norm* of a vector x in $\mathbb{R}^n$, and the *Euclidean metric* measuring the distance between two points x and y in $\mathbb{R}^n$. Each of these generalizes a familiar concept from $\mathbb{R}$. Namely, when $n = 1$, and x and y are just real numbers, the Euclidean inner product of x and y is just the product xy of the numbers x and y ; the Euclidean norm of x is simply the absolute value $|x|$ of x ; and the Euclidean distance between x and y is the absolute value $|x - y|$ of their difference.

Given $x, y \in \mathbb{R}^n$, the *Euclidean inner product* of the vectors x and y , denoted $x \cdot y$, is defined as:

$$x \cdot y = \sum_{i=1}^n x_i y_i.$$

We shall henceforth refer to the Euclidean inner product simply as the inner product.

Theorem 1.1 *The inner product has the following properties for any vectors $x, y, z \in \mathbb{R}^n$ and scalars $a, b \in \mathbb{R}$:*

1. *Symmetry:* $x \cdot y = y \cdot x$.
2. *Bilinearity:* $(ax + by) \cdot z = ax \cdot z + by \cdot z$ and $x \cdot (ay + bz) = x \cdot ay + x \cdot bz$.
3. *Positivity:* $x \cdot x \geq 0$, with equality holding if and only if $x = 0$.

Proof Symmetry and bilinearity are easy to verify from the definition of the inner product. To check that positivity holds, note that the square of a real number is always nonnegative, and can be zero if and only if the number is itself zero. It follows that as the sum of squared real numbers, $x \cdot x = \sum_{i=1}^n x_i^2$ is always nonnegative, and is zero if and only if $x_i = 0$ for each i , i.e., if and only if $x = 0$. $\square$

The inner product also satisfies a very useful condition called the *Cauchy–Schwartz inequality*:

Theorem 1.2 (Cauchy–Schwartz Inequality) *For any $x, y \in \mathbb{R}^n$ we have*

$$|x \cdot y| \leq (x \cdot x)^{1/2} (y \cdot y)^{1/2}.$$

Proof For notational ease, let $X = x \cdot x$, $Y = y \cdot y$, and $Z = x \cdot y$. Then, the result will be proved if we show that $XY \geq Z^2$, since the required inequality will follow simply by taking square roots on both sides.

If $x = 0$, then $Z = X = 0$, and the inequality holds trivially. Suppose, therefore, that $x \neq 0$. Note that by the positivity property of the inner product, we must then

have $X > 0$. The positivity property also implies that for any scalar $a \in \mathbb{R}$, we have

$$\begin{aligned} 0 &\leq (ax + y) \cdot (ax + y) \\ &= a^2 x \cdot x + 2ax \cdot y + y \cdot y \\ &= a^2 X + 2aZ + Y. \end{aligned}$$

In particular, this inequality must hold for $a = -Z/X$. When this value of a is used in the equation above, we obtain

$$\left(\frac{Z}{X}\right)^2 X - 2\left(\frac{Z}{X}\right)Z + Y = -\left(\frac{Z^2}{X}\right) + Y \geq 0,$$

or $Y \geq Z^2/X$. Since $X > 0$, this in turn implies $XY \geq Z^2$, as required. $\square$

The *Euclidean norm* (henceforth, simply the *norm*) of a vector $x \in \mathbb{R}^n$, denoted $\|x\|$, is defined as

$$\|x\| = \left(\sum_{i=1}^n x_i^2\right)^{1/2}.$$

The norm is related to the inner product through the identity

$$\|x\| = (x \cdot x)^{1/2}$$

for all $x \in \mathbb{R}^n$; in particular, the Cauchy–Schwartz inequality may be written as

$$|x \cdot y| \leq \|x\| \|y\|.$$

Our next result, which describes some useful properties of the norm, uses this observation.

Theorem 1.3 *The norm satisfies the following properties at all $x, y \in \mathbb{R}^n$, and $a \in \mathbb{R}$:*

1. *Positivity:* $\|x\| \geq 0$, with equality if and only if $x = 0$.
2. *Homogeneity:* $\|ax\| = |a| \cdot \|x\|$.
3. *Triangle Inequality:* $\|x + y\| \leq \|x\| + \|y\|$.

Proof The positivity property of the norm follows from the positivity property of the inner product, and the fact that $\|x\| = (x \cdot x)^{1/2}$. Homogeneity obtains since

$$\|ax\| = \left(\sum_{i=1}^n a^2 x_i^2\right)^{1/2} = \left(a^2 \sum_{i=1}^n x_i^2\right)^{1/2} = |a| \|x\|.$$

The triangle inequality is a little trickier; we will need the Cauchy–Schwartz inequality to establish it. Observe that for any x and y in $\mathbb{R}^n$, we have

$$\|x + y\|^2 = (x + y) \cdot (x + y) = \|x\|^2 + 2x \cdot y + \|y\|^2.$$

By the Cauchy–Schwartz inequality, $x \cdot y \leq \|x\| \|y\|$. Substituting this in the previous equation, we obtain

$$\|x + y\|^2 \leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

The proof is completed by taking square roots on both sides. $\square$

The *Euclidean distance* $d(x, y)$ between two vectors x and y in $\mathbb{R}^n$ is given by

$$d(x, y) = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{1/2}.$$

The distance function d is called a *metric*, and is related to the norm $\|\cdot\|$ through the identity

$$d(x, y) = \|x - y\|$$

for all $x, y \in \mathbb{R}^n$.

Theorem 1.4 *The metric d satisfies the following properties at all $x, y, z \in \mathbb{R}^n$:*

1. *Positivity:* $d(x, y) \geq 0$ with equality if and only if $x = y$.
2. *Symmetry:* $d(x, y) = d(y, x)$.
3. *Triangle Inequality:* $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in \mathbb{R}^n$.

Proof The positivity property of the metric follows from the positivity property of the norm, and the observation that $d(x, y) = \|x - y\|$. Symmetry is immediate from the definition. The inequality $d(x, z) \leq d(x, y) + d(y, z)$ is the same as

$$\|x - z\| \leq \|x - y\| + \|y - z\|.$$

This is just the triangle inequality for norms, which we have already established. $\square$

The concepts of inner product, norm, and metric can be defined on any abstract vector space, and not just $\mathbb{R}^n$. In fact, the properties we have listed in Theorems 1.1, 1.3, and 1.4 are, in abstract vector spaces, defining characteristics of the respective concepts. Thus, for instance, an inner product on a vector space is *defined* to be any operator on that space that satisfies the three properties of symmetry, bilinearity, and positivity; while a norm on that space is defined to be any operator that meets the conditions of positivity, homogeneity, and the triangle inequality. For more on this, see Appendix C.

1.2 Sets and Sequences in $\mathbb{R}^n$

1.2.1 Sequences and Limits

A *sequence* in $\mathbb{R}^n$ is the specification of a point $x_k \in \mathbb{R}^n$ for each integer $k \in \{1, 2, \dots\}$. The sequence is usually written as

$$x_1, x_2, x_3, \dots$$

or, more compactly, simply as $\{x_k\}$. Occasionally, where notational clarity will be enhanced by this change, we will use superscripts instead of subscripts, and denote the sequence by $\{x^k\}$.

A sequence of points $\{x_k\}$ in $\mathbb{R}^n$ is said to *converge* to a limit x (written $x_k \rightarrow x$) if the distance $d(x_k, x)$ between x_k and x tends to zero as k goes to infinity, i.e., if for all $\epsilon > 0$, there exists an integer $k(\epsilon)$ such that for all $k \geq k(\epsilon)$, we have $d(x_k, x) < \epsilon$. A sequence $\{x_k\}$ which converges to a limit x is called a *convergent sequence*.

For example, the sequence $\{x_k\}$ in $\mathbb{R}$ defined by $x_k = 1/k$ for all k is a convergent sequence, with limit $x = 0$. To see this, let any $\epsilon > 0$ be given. Let $k(\epsilon)$ be any integer such that $k(\epsilon) > 1/\epsilon$. Then, for all $k > k(\epsilon)$, we have $d(x_k, 0) = d(1/k, 0) = 1/k < 1/k(\epsilon) < \epsilon$, so indeed, $x_k \rightarrow 0$.

Theorem 1.5 *A sequence can have at most one limit. That is, if $\{x_k\}$ is a sequence in $\mathbb{R}^n$ converging to a point $x \in \mathbb{R}^n$, it cannot also converge to a point $y \in \mathbb{R}^n$ for $y \neq x$.*

Proof This follows from a simple application of the triangle inequality. If $x_k \rightarrow x$ and $y \neq x$, then

$$d(x_k, y) \geq d(x, y) - d(x_k, x).$$

Since $d(x, y) > 0$ and $d(x_k, x) \rightarrow 0$, this inequality shows that $d(x_k, y)$ cannot go to zero as k goes to infinity, so $x_k \rightarrow y$ is impossible. $\square$

A sequence $\{x_k\}$ in $\mathbb{R}^n$ is called a *bounded* sequence if there exists a real number M such that $\|x_k\| \leq M$ for all k . A sequence $\{x_k\}$ which is not bounded is said to be *unbounded*; that is, $\{x_k\}$ is an unbounded sequence if for any $M \in \mathbb{R}$, there exists $k(M)$ such that $\|x_{k(M)}\| > M$.

Theorem 1.6 *Every convergent sequence in $\mathbb{R}^n$ is bounded.*

Proof Suppose $x_k \rightarrow x$. Let $\epsilon = 1$ in the definition of convergence. Then, there exists $k(1)$ such that for all $k \geq k(1)$, $d(x_k, x) < 1$. Since $d(x_k, x) = \|x_k - x\|$, an

application of the triangle inequality yields for any $k \geq k(1)$

$$\begin{aligned}\|x_k\| &= \|(x_k - x) + x\| \\ &\leq \|x_k - x\| + \|x\| \\ &< 1 + \|x\|.\end{aligned}$$

Now define M to be the maximum of the finite set of numbers

$$\{\|x_1\|, \dots, \|x_{k(1)-1}\|, 1 + \|x\|\}.$$

Then, $M \geq \|x_k\|$ for all k , completing the proof. $\square$

While Theorem 1.5 established that a sequence can have at most one limit, Theorem 1.6 implies that a sequence may have *no* limit at all. Indeed, because every convergent sequence must be bounded, it follows that if $\{x_k\}$ is an unbounded sequence, then $\{x_k\}$ cannot converge. Thus, for instance, the sequence $\{x_k\}$ in $\mathbb{R}$ defined by $x_k = k$ for all k is a non-convergent sequence.¹

However, unboundedness is not the only reason a sequence may fail to converge. Consider the following example: let $\{x_k\}$ in $\mathbb{R}$ be given by

$$x_k = \begin{cases} \frac{1}{k}, & k = 1, 3, 5, \dots \\ 1 - \frac{1}{k}, & k = 2, 4, 6, \dots \end{cases}$$

This sequence is bounded since we have $|x_k| \leq 1$ for all k . However, it does not possess a limit. The reason here is that the odd terms of the sequence are converging to the point 0, while the even terms are converging to the point 1. Since a sequence can have only one limit, this sequence does not converge.

Our next result shows that convergence of a sequence $\{x^k\}$ in $\mathbb{R}^n$ is equivalent to convergence in each coordinate. This gives us an alternative way to establish convergence in $\mathbb{R}^n$. We use superscripts to denote the sequence in this result to avoid confusion between the k -th element x_k of the sequence, and the i -th coordinate x_i of a vector x .

Theorem 1.7 *A sequence $\{x^k\}$ in $\mathbb{R}^n$ converges to a limit x if and only if $x_i^k \rightarrow x_i$ for each $i \in \{1, \dots, n\}$, where $x^k = (x_1^k, \dots, x_n^k)$ and $x = (x_1, \dots, x_n)$.*

¹This may also be shown directly: for any fixed candidate limit x the distance $d(x_k, x) = |x - x_k| = |x - k|$ becomes unbounded as k goes to infinity. It follows that no $x \in \mathbb{R}$ can be a limit of this sequence, and therefore that it does not possess a limit.

Proof We will use the fact that the Euclidean distance between two points $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbb{R}^n$ can be written as

$$d(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^2 \right)^{1/2},$$

where $|x_i - y_i|$ is the Euclidean distance between x_i and y_i in $\mathbb{R}$.

First, suppose that $x^k \rightarrow x$. We are to show that $x_i^k \rightarrow x_i$ for each i , i.e., that, given any i and $\epsilon > 0$, there exists $k_i(\epsilon)$ such that for $k \geq k_i(\epsilon)$, we have $|x_i^k - x_i| < \epsilon$.

So let $\epsilon > 0$ be given. By definition of $x^k \rightarrow x$, we know that there exists $k(\epsilon)$ such that $d(x^k, x) < \epsilon$ for all $k \geq k(\epsilon)$. Therefore, for $k \geq k(\epsilon)$ and any i , we obtain:

$$|x_i^k - x_i| = \left(|x_i^k - x_i|^2 \right)^{1/2} \leq \left(\sum_{j=1}^n |x_j^k - x_j|^2 \right)^{1/2} = d(x^k, x) < \epsilon.$$

Setting $k_i(\epsilon) = k(\epsilon)$ for each i , the proof that $x_i^k \rightarrow x_i$ for each i is complete.

Now, suppose that $\{x_i^k\}$ converges to x_i for each i . Let $\epsilon > 0$ be given. We will show that there is $k(\epsilon)$ such that $d(x^k, x) < \epsilon$ for all $k \geq k(\epsilon)$, which will establish that $x^k \rightarrow x$.

Define $\eta = \epsilon/\sqrt{n}$. For each i , there exists $k_i(\eta)$ such that for $k \geq k_i(\eta)$, we have $|x_i^k - x_i| < \eta$. Define $k(\epsilon)$ to be the maximum of the finite set of numbers $k_1(\eta), \dots, k_n(\eta)$. Then, for $k \geq k(\epsilon)$, we have $|x_i^k - x_i| < \eta$ for all i , so

$$d(x^k, x) = \left(\sum_{i=1}^n |x_i^k - x_i|^2 \right)^{1/2} < \left(\sum_{i=1}^n \left[\frac{\epsilon}{\sqrt{n}} \right]^2 \right)^{1/2} = \epsilon,$$

which completes the proof. $\square$

Theorem 1.7 makes it easy to prove the following useful result:

Theorem 1.8 *Let $\{x^k\}$ be a sequence in $\mathbb{R}^n$ converging to a limit x . Suppose that for every k , we have $a \leq x^k \leq b$, where $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$ are some fixed vectors in $\mathbb{R}^n$. Then, it is also the case that $a \leq x \leq b$.*

Proof The theorem will be proved if we show that $a_i \leq x_i \leq b_i$ for each $i \in \{1, \dots, n\}$. Suppose that the result were false, so for some i , we had $x_i < a_i$ (say). Since $x^k \rightarrow x$, it is the case by Theorem 1.7 that $x_j^k \rightarrow x_j$ for each $j \in \{1, \dots, n\}$; in particular, $x_i^k \rightarrow x_i$. But $x_i^k \rightarrow x_i$ combined with $x_i < a_i$ implies that for all large k , we must have $x_i^k < a_i$. This contradicts the hypothesis that $a_i \leq x_i^k \leq b_i$ for all k . A similar argument establishes that $x_i > b_i$ also leads to a contradiction. Thus, $a_i \leq x_i \leq b_i$, and the proof is complete. $\square$

1.2.2 Subsequences and Limit Points

Let a sequence $\{x_k\}$ in $\mathbb{R}^n$ be given. Let m be any rule that assigns to each $k \in \mathbb{N}$ a value $m(k) \in \mathbb{N}$. Suppose further that m is *increasing*, i.e., for each $k \in \mathbb{N}$, we have $m(k) < m(k+1)$. Given $\{x_k\}$, we can now define a new sequence $\{x_{m(k)}\}$, whose k -th element is the $m(k)$ -th element of the sequence $\{x_k\}$. This new sequence is called a *subsequence* of $\{x_k\}$. Put differently, a subsequence of a sequence is any infinite subset of the original sequence that preserves the ordering of terms.

Even if a sequence $\{x_k\}$ is not convergent, it may contain subsequences that converge. For instance, the sequence $0, 1, 0, 1, 0, 1, \dots$ has no limit, but the subsequences $0, 0, 0, \dots$ and $1, 1, 1, \dots$ which are obtained from the original sequence by selecting the odd and even elements, respectively, are both convergent.

If a sequence contains a convergent subsequence, the limit of the convergent subsequence is called a *limit point* of the original sequence. Thus, the sequence $0, 1, 0, 1, 0, 1, \dots$ has two limit points 0 and 1. The following result is simply a restatement of the definition of a limit point:

Theorem 1.9 *A point x is a limit point of the sequence $\{x_k\}$ if and only if for any $\epsilon > 0$, there are infinitely many indices m for which $d(x, x_m) < \epsilon$.*

Proof If x is a limit point of $\{x_k\}$ then there must be a subsequence $\{x_{m(k)}\}$ that converges to x . By definition of convergence, it is the case that for any $\epsilon > 0$, all but finitely many elements of the sequence $\{x_{m(k)}\}$ must be within ϵ of x . Therefore, infinitely many elements of the sequence $\{x_k\}$ must also be within ϵ of x .

Conversely, suppose that for every $\epsilon > 0$, there are infinitely many m such that $d(x_m, x) < \epsilon$. Define a subsequence $\{x_{m(k)}\}$ as follows: let $m(1)$ be any m for which $d(x_m, x) < 1$. Now for $k = 2, 3, \dots$ define successively $m(k)$ to be any m that satisfies (a) $d(x, x_m) < 1/k$, and (b) $m > m(k-1)$. This construction is feasible, since for each k , there are infinitely many m satisfying $d(x_m, x) < 1/k$. Moreover, the sequence $\{x_{m(k)}\}$ evidently converges to x , so x is a limit point of $\{x_k\}$. $\square$

If a sequence $\{x_k\}$ is convergent (say, to a limit x), then it is apparent that every subsequence of $\{x_k\}$ must converge to x . It is less obvious, but also true, that if *every* subsequence $\{x_{m(k)}\}$ of a given sequence $\{x_k\}$ converges to the limit x , then $\{x_k\}$ itself converges to x . We do not offer a proof of this fact here, since it may be easily derived as a consequence of other considerations. See Corollary 1.19 below.

In general, a sequence $\{x_k\}$ may have any number of limit points. For instance, *every* positive integer arises as a limit point of the sequence

$$1, 1, 2, 1, 2, 3, 1, 2, 3, 4, \dots$$

Of course, it is also possible that *no* subsequence of a given sequence converges, so a given sequence may have no limit points at all. A simple example is the sequence $\{x_k\}$ in $\mathbb{R}$ defined by $x_k = k$ for all k : every subsequence of this sequence diverges to $+\infty$.

1.2.3 Cauchy Sequences and Completeness

A sequence $\{x_k\}$ in $\mathbb{R}^n$ is said to satisfy the *Cauchy criterion* if for all $\epsilon > 0$ there is an integer $k(\epsilon)$ such that for all $m, l \geq k(\epsilon)$, we have $d(x_m, x_l) < \epsilon$. Informally, a sequence $\{x_k\}$ satisfies the Cauchy criterion if, by choosing k large enough, the distance between any two elements x_m and x_l in the “tail” of the sequence

$$x_k, x_{k+1}, x_{k+2}, \dots$$

can be made as small as desired. A sequence which satisfies the Cauchy criterion is called a *Cauchy sequence*.

An example of a Cauchy sequence is given by the sequence $\{x_k\}$ in $\mathbb{R}$ defined by $x_k = 1/k^2$ for all k . To check that this sequence does, in fact, satisfy the Cauchy criterion, let any $\epsilon > 0$ be given. Let $k(\epsilon)$ be any integer k that satisfies $k^2\epsilon > 2$. For $m, l \geq k(\epsilon)$, we have

$$d(x_m, x_l) = \left| \frac{1}{m^2} - \frac{1}{l^2} \right| \leq \left| \frac{1}{m^2} + \frac{1}{l^2} \right| \leq \left| \frac{1}{[k(\epsilon)]^2} + \frac{1}{[k(\epsilon)]^2} \right| = \frac{2}{[k(\epsilon)]^2},$$

and this last term is less than ϵ by choice of $k(\epsilon)$.

Our first result deals with the analog of Theorem 1.7 for Cauchy sequences. It establishes that a sequence in $\mathbb{R}^n$ is a Cauchy sequence if and only if each of the coordinate sequences is a Cauchy sequence in $\mathbb{R}$.

Theorem 1.10 *A sequence $\{x^k\}$ in $\mathbb{R}^n$ is a Cauchy sequence if and only if for each $i \in \{1, \dots, n\}$, the sequence $\{x_i^k\}$ is a Cauchy sequence in $\mathbb{R}$.*

Proof Let $\{x^k\}$ be a Cauchy sequence in $\mathbb{R}^n$. We are to show that for any i and any $\epsilon > 0$, there is $k_i(\epsilon)$ such that for all $m, l \geq k_i(\epsilon)$, we have $|x_i^m - x_i^l| < \epsilon$. So let $\epsilon > 0$ and i be given. Since $\{x^k\}$ is a Cauchy sequence, there is $k(\epsilon)$ such that for all $m, l \geq k(\epsilon)$, we have $d(x^m, x^l) < \epsilon$. Therefore, for any $m, l \geq k(\epsilon)$, we have

$$|x_i^m - x_i^l| = (|x_i^m - x_i^l|^2)^{1/2} \leq \left(\sum_{j=1}^n |x_j^m - x_j^l|^2 \right)^{1/2} = d(x^m, x^l) < \epsilon.$$

By setting $k_i(\epsilon) = k(\epsilon)$, the proof that $\{x_i^k\}$ is a Cauchy sequence for each i is complete.

Now suppose $\{x_i^k\}$ is a Cauchy sequence for each i . We are to show that for any $\epsilon > 0$, there is $k(\epsilon)$ such that $m, l \geq k(\epsilon)$ implies $d(x^m, x^l) < \epsilon$. So let $\epsilon > 0$ be given. Define $\eta = \epsilon/\sqrt{n}$. Since each $\{x_i^k\}$ is a Cauchy sequence, there exists $k_i(\eta)$ such that for all $m, l \geq k_i(\eta)$, we have $|x_i^m - x_i^l| < \eta$. Define $k(\epsilon) = \max\{k_1(\eta), \dots, k_n(\eta)\}$. Then, for $m, l \geq k(\epsilon)$, we have

$$d(x^m, x^l) = \left(\sum_{i=1}^n |x_i^m - x_i^l|^2 \right)^{1/2} < \left(\sum_{i=1}^n \left[\frac{\epsilon}{\sqrt{n}} \right]^2 \right)^{1/2} = \epsilon,$$

and the proof is complete. $\square$

Observe that the important difference between the definition of a convergent sequence and that of a Cauchy sequence is that the limit of the sequence is explicitly involved in the former, but plays no role in the latter. Our next result shows, however, that the two concepts are intimately linked:

Theorem 1.11 *A sequence $\{x_k\}$ in $\mathbb{R}^n$ is a Cauchy sequence if and only if it is a convergent sequence, i.e., if and only if there is $x \in \mathbb{R}^n$ such that $x_k \rightarrow x$.*

Proof The proof that every convergent sequence must also be a Cauchy sequence is simple. Suppose $x_k \rightarrow x$. Let $\epsilon > 0$ be given. Define $\eta = \epsilon/2$. Since $x_k \rightarrow x$, there exists $k(\eta)$ such that for all $j \geq k(\eta)$, we have $d(x_j, x) < \eta$. It follows by using the triangle inequality that for all $j, l \geq k(\eta)$, we have

$$d(x_j, x_l) \leq d(x_j, x) + d(x_l, x) < \eta + \eta = \epsilon.$$

Thus, setting $k(\epsilon) = k(\eta)$, the proof that $\{x_k\}$ is a Cauchy sequence is complete.

The proof that every Cauchy sequence must converge unfortunately requires more apparatus than we have built up. In particular, it requires a formal definition of the notion of a “real number,” and the properties of real numbers. Appendix B, which presents such a formal description, proves that every Cauchy sequence in $\mathbb{R}$ must converge. An appeal to Theorem 1.10 then completes the proof. $\square$

Any metric space which has the property that every Cauchy sequence is also a convergent sequence is said to be *complete*. Thus, by Theorem 1.11, $\mathbb{R}^n$ is complete. It is important to note that not all metric spaces are complete (so the definition is not vacuous). For more on this, see Appendix C.

Even without appealing to the completeness of $\mathbb{R}^n$, it is possible to show that a Cauchy sequence must possess two properties that all convergent sequences must have: namely, that it must be bounded, and that it has at most one limit point.

Theorem 1.12 Let $\{x_k\}$ be a Cauchy sequence in $\mathbb{R}^n$. Then,

1. $\{x_k\}$ is bounded.
2. $\{x_k\}$ has at most one limit point.

Proof To see that every Cauchy sequence $\{x_k\}$ in $\mathbb{R}^n$ must be bounded, take $\epsilon = 1$ in the Cauchy criterion. Then, there exists an integer $k(1)$ such that for all $j, l \geq k(1)$, we have $d(x_j, x_l) < 1$. An application of the triangle inequality now implies for all $j > k(1)$ that

$$\begin{aligned}\|x_j\| &= \|(x_j - x_{k(1)}) + x_{k(1)}\| \\ &\leq \|x_j - x_{k(1)}\| + \|x_{k(1)}\| \\ &< 1 + \|x_{k(1)}\|.\end{aligned}$$

Let M be the maximum of the finite set of numbers

$$\{\|x_1\|, \dots, \|x_{k(1)-1}\|, 1 + \|x_{k(1)}\|\}.$$

Then, by construction, $M \geq \|x_k\|$ for all k , and the proof is complete.

To see that a Cauchy sequence cannot have two or more limit points, suppose that x is a limit point of the Cauchy sequence $\{x_k\}$. We will show that $x_k \rightarrow x$. Let $\epsilon > 0$ be given, and let $\eta = \epsilon/2$. Since $\{x_k\}$ is Cauchy, there exists $k(\eta)$ such that $d(x_j, x_l) < \eta$ for all $j, l \geq k(\eta)$. Moreover, since x is a limit point of $\{x_k\}$, there are elements of the sequence $\{x_k\}$ that lie arbitrarily close to x ; in particular, we can find $m \geq k(\eta)$ such that $d(x_m, x) < \eta$. Therefore, if $j \geq k(\eta)$,

$$\begin{aligned}d(x_j, x) &\leq d(x_j, x_m) + d(x_m, x) \\ &\leq \eta + \eta \\ &= \epsilon.\end{aligned}$$

Since $\epsilon > 0$ was arbitrary, this string of inequalities shows precisely that $x_k \rightarrow x$. $\square$

Finally, two Cauchy sequences $\{x_k\}$ and $\{y_k\}$ are said to be *equivalent* if for all $\epsilon > 0$, there is $k(\epsilon)$ such that for all $j \geq k(\epsilon)$ we have $d(x_j, y_j) < \epsilon$. We write this as $\{x_k\} \sim \{y_k\}$. It is easy to see that $\sim$ is, in fact, an equivalence relationship. It is reflexive ($\{x_k\} \sim \{x_k\}$) and symmetric ($\{x_k\} \sim \{y_k\}$ implies $\{y_k\} \sim \{x_k\}$). It is also transitive: $\{x_k\} \sim \{y_k\}$ and $\{y_k\} \sim \{z_k\}$ implies $\{x_k\} \sim \{z_k\}$. To see this, let $\epsilon > 0$ be given. Let $\eta = \epsilon/2$. Since $\{x_k\} \sim \{y_k\}$, there is $k_1(\eta)$ such that for all $k \geq k_1(\eta)$, we have $d(x_k, y_k) < \eta$. Similarly, there is $k_2(\eta)$ such that for all $k \geq k_2(\eta)$, $d(y_k, z_k) < \eta$. Let $k(\epsilon) = \max\{k_1(\eta), k_2(\eta)\}$. Then, for $k \geq k(\epsilon)$, we have $d(x_k, z_k) \leq d(x_k, y_k) + d(y_k, z_k) < \eta + \eta = \epsilon$, and we indeed have $\{x_k\} \sim \{z_k\}$.

Equivalent Cauchy sequences must have the same limit.² For, suppose $\{x_k\} \sim \{y_k\}$ and $x_k \rightarrow x$. Given $\epsilon > 0$, there is $k_1(\epsilon)$ such that $k \geq k_1(\epsilon)$ implies $d(x_k, x) < \epsilon/2$. There is also $k_2(\epsilon)$ such that $k \geq k_2(\epsilon)$ implies $d(x_k, y_k) < \epsilon/2$. Therefore, letting $k(\epsilon) = \max\{k_1(\epsilon), k_2(\epsilon)\}$, $k \geq k(\epsilon)$ implies $d(x, y_k) \leq d(x, x_k) + d(x_k, y_k) < \epsilon/2 + \epsilon/2 = \epsilon$, which states precisely that $y_k \rightarrow x$.

1.2.4 Suprema, Infima, Maxima, Minima

Let A be a nonempty subset of $\mathbb{R}$. The set of *upper bounds* of A , denoted $U(A)$, is defined as

$$U(A) = \{u \in \mathbb{R} \mid u \geq a \text{ for all } a \in A\}$$

while the set of *lower bounds* of A , denoted $L(A)$, is given by

$$L(A) = \{l \in \mathbb{R} \mid l \leq a \text{ for all } a \in A\}.$$

In general, $U(A)$ and/or $L(A)$ could be empty. For instance, if $A = \mathbb{N}$, the set of positive integers, then $U(A)$ is empty; if $A = \mathbb{Z}$, the set of all integers, then both $U(A)$ and $L(A)$ are empty. If $U(A)$ is nonempty, then A is said to be *bounded above*. If $L(A)$ is nonempty, then A is said to be *bounded below*.

The *supremum* of A , written $\sup A$, is defined to be the *least upper bound* of A . Namely, if $U(A)$ is nonempty, then $\sup A$ is defined to be the unique point $a^* \in U(A)$ such that $a^* \leq u$ for all $u \in U(A)$. If $U(A)$ is empty, on the other hand, then A has no finite upper bounds, so, by convention, we set $\sup A = +\infty$.

Similarly, the *infimum* of A , denoted $\inf A$, is defined to be the *greatest lower bound* of A . That is, when $L(A)$ is nonempty, then $\inf A$ is the unique point $\hat{a} \in L(A)$ such that $\hat{a} \geq l$ for all $l \in L(A)$. If $L(A)$ is empty, then A admits no finite lower bounds, so, by convention, we set $\inf A = -\infty$.

Theorem 1.13 *If $U(A)$ is nonempty, the supremum of A is well defined, i.e., there is $a^* \in U(A)$ such that $a^* \leq u$ for all $u \in U(A)$. Similarly, if $L(A)$ is nonempty, the infimum of A is well defined, i.e., there is $\hat{a} \in L(A)$ such that $\hat{a} \geq l$ for all $l \in L(A)$.*

Remark 1 By our conventions that $\sup A = +\infty$ when $U(A)$ is empty and $\inf A = -\infty$ when $L(A)$ is empty, this will establish that $\sup A$ and $\inf A$ are defined for any nonempty set $A \subset \mathbb{R}$. □

Remark 2 To avoid legitimate confusion, we should stress the point that some

²In fact, the procedure of constructing the real numbers from the rational numbers using Cauchy sequences of rationals (see Appendix B) *defines* a real number to be an equivalence class of Cauchy sequences of rational numbers.

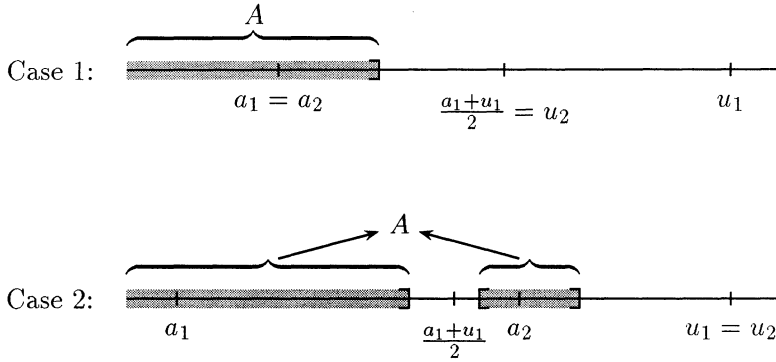


Fig. 1.2. Constructing the Supremum

authors (e.g., Apostol, 1967, or Bartle, 1964) take an axiomatic approach to the real line. In this approach, Theorem 1.13 is an *axiom*, indeed, a key axiom. Other authors adopt a constructive approach to the real line, building the real numbers from the rationals by using, for example, the method of Dedekind cuts (Rudin, 1976), or equivalent Cauchy sequences of rationals (Hewitt and Stromberg, 1965; Strichartz, 1982; Appendix B in this book). In this case, the result that bounded sets have well-defined suprema is genuinely a “theorem.” We implicitly adopt the latter approach in this chapter. $\square$

Proof We prove that $\sup A$ is well defined whenever A is bounded above. A similar procedure can be employed to show that $\inf A$ is well-defined whenever A is bounded below. The details are left as an exercise.

So suppose A is bounded above and $U(A)$ is nonempty. We will construct two sequences $\{a_k\}$ and $\{u_k\}$ that have a common limit a^* . The sequence $\{a_k\}$ will consist entirely of points from A and will converge “upwards” to a^* , while the sequence $\{u_k\}$ will consist solely of points from $U(A)$, and will converge “downwards” to a^* . It will follow quite easily from the construction of these sequences that the common limit a^* is, in fact, $\sup A$.

The required sequences are constructed using a divide-and-conquer procedure. Pick any point a_1 in A and any point u_1 in $U(A)$. Let $z_1 = (a_1 + u_1)/2$ be their midpoint. Of course, $a_1 \leq z_1 \leq u_1$. There are two possible cases: $z_1 \in U(A)$ and $z_1 \notin U(A)$. In case 1, where $z_1 \in U(A)$, set $a_2 = a_1$ and $u_2 = z_1$. In case 2, where $z_1 \notin U(A)$, there must exist some point $a \in A$ such that $a \geq z_1$.³ In this case, let $a_2 = a$ and $u_2 = u_1$ (see Figure 1.2).

³Note that $z_1 \notin U(A)$ does *not* imply that $z_1 \in A$. For instance, if $A = [0, 1] \cup [3, 4]$ and $z_1 = 2$, then $z_1 \notin U(A)$ and $z_1 \notin A$.

Note that in either case we have $a_2 \in A$ and $u_2 \in U(A)$. Moreover, we must also have $a_1 \leq a_2$ and $u_1 \geq u_2$. Finally, in the first case, we have $d(a_2, u_2) = d(a_1, (a_1 + u_1)/2) = d(a_1, u_1)/2$, while in the second case, we have $d(a_2, u_2) = d(a, u_1) \leq d(a_1, u_1)/2$, since $a \geq (a_1 + u_1)/2$. Thus, in either case, we have $d(a_2, u_2) \leq d(a_1, u_1)/2$.

We iterate on this procedure. Let $z_2 = (a_2 + u_2)/2$ be the midpoint of a_2 and u_2 . If $z_2 \in U(A)$, let $u_3 = z_2$ and $a_3 = a_2$. If $z_2 \notin U(A)$, then there must exist $a' \in A$ such that $a' \geq z_2$; in this case, set $a_3 = a'$ and $u_3 = u_2$. Then, it is once again true that the following three conditions hold in either case: first, we have $a_3 \in A$ and $u_3 \in U(A)$; second, $a_3 \geq a_2$ and $u_3 \leq u_2$; finally, $d(a_3, u_3) \leq d(a_2, u_2)/2 \leq d(a_1, u_1)/4$.

Continuing in this vein, we obtain sequences $\{a_k\}$ and $\{u_k\}$ such that

1. for each k , $a_k \in A$ and $u_k \in U(A)$;
2. $\{a_k\}$ is a nondecreasing sequence, and $\{u_k\}$ is a nonincreasing sequence; and
3. $d(a_k, u_k) \leq d(a_1, u_1)/2^k$.

Property 2 implies in particular that $\{a_k\}$ and $\{u_k\}$ are bounded sequences. It now follows from property 3 that $\{a_k\}$ and $\{u_k\}$ are equivalent Cauchy sequences, and, therefore, that they have the same limit a^* . We will show that $a^* = \sup A$, by showing first that $a^* \in U(A)$, and then that $a^* \leq u$ for all $u \in U(A)$.

Pick any $a \in A$. Since $u_k \in U(A)$ for each k , we have $u_k \geq a$ for each k , so $a^* = \lim_k u_k \geq a$. Since $a \in A$ was arbitrary, this inequality implies $a^* \in U(A)$. Now pick any $u \in U(A)$. Since $a_k \in A$ for each k , we must have $u \geq a_k$ for each k . Therefore, we must also have $u \geq \lim_k a_k = a^*$. Since $u \in U(A)$ was arbitrary, this inequality implies $u \geq a^*$ for all $u \in U(A)$.

Summing up, $a^* \in U(A)$ and $a^* \leq u$ for all $u \in U(A)$. By definition, this means $a^* = \sup A$, and the proof is complete. $\square$

The following result is an immediate consequence of the definition of the supremum. We raise it to the level of a theorem, since it is an observation that comes in handy quite frequently:

Theorem 1.14 *Suppose $\sup A$ is finite. Then, for any $\epsilon > 0$, there is $a(\epsilon) \in A$ such that $a(\epsilon) > \sup A - \epsilon$.*

Proof For notational ease, let $a^* = \sup A$. Suppose the theorem failed for some $\epsilon > 0$, that is, suppose there were $\epsilon > 0$ such that $a \leq a^* - \epsilon$ for all $a \in A$. Then, $a^* - \epsilon$ would be an upper bound of A . But this violates the definition of a^* as the least upper bound, since $a^* - \epsilon$ is obviously strictly smaller than a^* . $\square$

A similar result to Theorem 1.14 evidently holds for the infimum. It is left to the reader as an exercise to fill in the details.

Two concepts closely related to the supremum and the infimum are the *maximum* and the *minimum* of a nonempty set $A \subset \mathbb{R}$. The *maximum* of A , denoted $\max A$, is defined as a point $z \in A$ such that $z \geq a$ for all $a \in A$. The *minimum* of A , denoted $\min A$, is defined as a point $w \in A$ such that $w \leq a$ for all $a \in A$. By definition, the maximum must be an upper bound of A , and the minimum must be a lower bound of A . Therefore, we can equivalently define $\max A = A \cap U(A)$, and $\min A = A \cap L(A)$.

It is very important to point out that while $\sup A$ and $\inf A$ are always defined for any nonempty set A (they could be infinite), $A \cap U(A)$ and $A \cap L(A)$ could both be empty, so $\max A$ and $\min A$ need *not* always exist. This is true even if $\sup A$ and $\inf A$ are both finite. For instance, if $A = (0, 1)$, we have $U(A) = \{x \mid x \geq 1\}$ and $L(A) = \{x \mid x \leq 0\}$, so $\sup A = 1$ and $\inf A = 0$, but $\max A$ and $\min A$ do not exist. Indeed, it follows from the definition that if $\max A$ (resp. $\min A$) is well defined, we must have $\max A = \sup A$ (resp. $\inf A = \min A$), so that $\max A$ exists if, and only if, $\sup A \in A$ (resp. $\min A$ exists if and only if $\inf A \in A$).

1.2.5 Monotone Sequences in $\mathbb{R}$

A sequence $\{x_k\}$ in $\mathbb{R}$ is said to be a *monotone increasing* sequence if it is the case that

$$x_{k+1} \geq x_k \text{ for all } k.$$

It is *monotone decreasing* if

$$x_{k+1} \leq x_k \text{ for all } k.$$

We will also refer to monotone increasing sequences as *nondecreasing* sequences, and to monotone decreasing sequences as *nonincreasing* sequences.

Monotone sequences possess a particularly simple asymptotic (i.e., limiting) structure, and this is one of the reasons they are of special interest. To state the formal result requires one more definition.

Say that a sequence $\{x_k\}$ in $\mathbb{R}$ *diverges to* $+\infty$ (written $x_k \uparrow +\infty$) if for all positive integers $p \in \mathbb{N}$, there is $k(p)$ such that for all $k \geq k(p)$, we have $x_k \geq p$; and that $\{x_k\}$ *diverges to* $-\infty$ (written $x_k \downarrow -\infty$) if for any positive integer $p \in \mathbb{N}$, there exists $k(p)$ such that for all $k \geq k(p)$, we have $x_k \leq -p$.

Observe that while a sequence that diverges to $\pm\infty$ must necessarily be unbounded, the converse is not always true: the sequence $\{x_k\}$ defined by

$$x_k = \begin{cases} 1, & \text{if } k \text{ is odd} \\ k, & \text{if } k \text{ is even} \end{cases}$$

is an unbounded sequence but it does not diverge to $+\infty$ (why?). On the other hand, it

is true that if $\{x_k\}$ is an unbounded sequence, it must contain at least one subsequence that diverges (to either $+\infty$ or $-\infty$).

The following result classifies the asymptotic behavior of monotone increasing sequences.

Theorem 1.15 *Let $\{x_k\}$ be a monotone increasing sequence in $\mathbb{R}$. If $\{x_k\}$ is unbounded, it must diverge to $+\infty$. If $\{x_k\}$ is bounded, it must converge to the limit x , where x is the supremum of the set of points $\{x_1, x_2, \dots\}$.*

Proof First suppose that $\{x_k\}$ is an unbounded sequence. Then, for any $p \in \mathbb{R}$, there exists an integer $k(p)$ such that $x_{k(p)} \geq p$. Since $\{x_k\}$ is monotone increasing, it is now the case that for any $k \geq k(p)$, we have $x_k \geq x_{k(p)} \geq p$. This says precisely that $\{x_k\}$ diverges to $+\infty$.

Next suppose that $\{x_k\}$ is a bounded sequence. We will show that $x_k \rightarrow x$, where, as defined in the statement of the theorem, x is the supremum of the set of points $\{x_1, x_2, \dots\}$. Let $\epsilon > 0$ be given. The proof will be complete if we show that there is $k(\epsilon)$ such that $d(x, x_k) < \epsilon$ for all $k \geq k(\epsilon)$.

Since x is the *least* upper bound of the set of points $\{x_1, x_2, \dots\}$, $x - \epsilon$ is not an upper bound of this set. Therefore, there exists $k(\epsilon)$ such that $x_{k(\epsilon)} > x - \epsilon$. Since $\{x_k\}$ is monotone increasing, it follows that $x_k > x - \epsilon$ for all $k \geq k(\epsilon)$. On the other hand, since x is an upper bound, it is certainly true that $x \geq x_k$ for all k . Combining these statements, we have

$$x_k \in (x - \epsilon, x], \quad k \geq k(\epsilon),$$

which, of course, implies that $d(x_k, x) < \epsilon$ for all $k \geq k(\epsilon)$. □

Since a sequence $\{x_k\}$ in $\mathbb{R}$ is monotone increasing if and only if $\{-x_k\}$ is monotone decreasing, Theorem 1.15 has the following immediate corollary:

Corollary 1.16 *Let $\{x_k\}$ be a monotone decreasing sequence in $\mathbb{R}$. If $\{x_k\}$ is unbounded, it must diverge to $-\infty$. If $\{x_k\}$ is bounded, it must converge to the limit x , where x is the infimum of the set of points $\{x_1, x_2, \dots\}$.*

1.2.6 The Lim Sup and Lim Inf

We now define the important concepts of the “lim sup” and the “lim inf” of a real-valued sequence. Throughout this discussion, we will assume as a convention that the values $\pm\infty$ are allowed as “limit points” of a sequence of real numbers $\{x_k\}$ in the following sense: the sequence $\{x_k\}$ will be said to have the limit point $+\infty$ if

$\{x_k\}$ contains a subsequence $\{x_{m(k)}\}$ that diverges to $+\infty$, and to have the limit point $-\infty$ if $\{x_k\}$ contains a subsequence $\{x_{l(k)}\}$ that diverges to $-\infty$. In particular, if the sequence $\{x_k\}$ itself diverges to $+\infty$ (resp. $-\infty$), then we will refer to $+\infty$ (resp. $-\infty$) as the limit of the sequence $\{x_k\}$.

So let a sequence $\{x_k\}$ in $\mathbb{R}$ be given. The *lim sup* of the sequence $\{x_k\}$ is then defined as the limit as $k \rightarrow \infty$ of the sequence $\{a_k\}$ defined by

$$a_k = \sup\{x_k, x_{k+1}, x_{k+2}, \dots\}$$

and is usually abbreviated as $\lim_{k \rightarrow \infty} \sup_{l \geq k} x_l$, or in more compact notation, as simply $\limsup_{k \rightarrow \infty} x_k$. To see that the *lim sup* is always well defined, note that there are only three possibilities that arise concerning the sequence $\{a_k\}$. First, it could be that $a_k = +\infty$ for some k , in which case $a_k = +\infty$ for all k . If this case is ruled out (so a_k is finite for each k), the sequence $\{a_k\}$ must satisfy $a_{k+1} \leq a_k$ for all k , since a_{k+1} is the supremum over a smaller set; that is, $\{a_k\}$ must be a *nonincreasing* sequence in $\mathbb{R}$. By Corollary 1.16, this specifies the only two remaining possibilities: either $\{a_k\}$ is unbounded, in which case $a_k \downarrow -\infty$, or $\{a_k\}$ is bounded, in which case $\{a_k\}$ converges to a limit a in $\mathbb{R}$. In all three cases, therefore, the limit of a_k is well defined (given our convention that $\pm\infty$ may be limiting values), and this limit is, of course, $\limsup_{k \rightarrow \infty} x_k$. Thus, the *lim sup* of *any* real-valued sequence always exists.

In a similar vein, the *lim inf* of the sequence $\{x_k\}$ is defined as the limit as $k \rightarrow \infty$ of $\{b_k\}$, where

$$b_k = \inf\{x_k, x_{k+1}, x_{k+2}, \dots\},$$

and is usually denoted $\lim_{k \rightarrow \infty} \inf_{l \geq k} x_l$, or just $\liminf_{k \rightarrow \infty} x_k$. Once again, either $b_k = -\infty$ for some k (in which case $b_k = -\infty$ for all k), or $\{b_k\}$ is a *nondecreasing* sequence in $\mathbb{R}$. Therefore, $\liminf_{k \rightarrow \infty} x_k$, is always well defined for any real-valued sequence $\{x_k\}$, although it could also take on infinite values.

The following result establishes two important facts: first, that the *lim sup* and *lim inf* of a sequence $\{x_k\}$ are themselves limit points of the sequence $\{x_k\}$, and second, that the *lim sup* is actually the supremum of the set of limit points of $\{x_k\}$, and the *lim inf* is the infimum of this set. This second part gives us an alternative interpretation of the *lim sup* and *lim inf*.

Theorem 1.17 *Let $\{x_k\}$ be a real-valued sequence, and let A denote the set of all limit points of $\{x_k\}$ (including $\pm\infty$ if $\{x_k\}$ contains such divergent subsequences). For notational convenience, let $a = \limsup_{k \rightarrow \infty} x_k$ and $b = \liminf_{k \rightarrow \infty} x_k$. Then:*

1. *There exist subsequences $m(k)$ and $l(k)$ of k such that $x_{m(k)} \rightarrow a$ and $x_{l(k)} \rightarrow b$.*
2. *$a = \sup A$ and $b = \inf A$.*